

Samar Mohanty

AI & ML engineer

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PROFILE SUMMARY

- AI/ML Engineer with 4 years of experience applying AI/ML to real-world business use cases across different industries.
- Comfortable owning a project end to end, from understanding the data and building the model through to deployment and monitoring in production.
- Comfortable across time-series forecasting, computer vision, and generative AI, with a recent shift toward LLM based systems including fine tuning, RAG and agentic workflows.
- Has applied these methods across supply chain, retail, and industrial inspection, adapting the approach to what each problem actually needs rather than forcing a single toolkit.

PROFESSIONAL EXPERIENCE

ML Ops engineer II

07/2026 – Present

Big Bang Boom solutions 🌐

Description:

Vision-based autonomous drone system for inspecting large structures in GPS-denied environments, using only camera input to navigate, capture, and reconstruct a site into a measurable 3D digital twin. The platform supports inspection use cases such as tunnel mapping, structural surveys of bridges and ships, and fault detection, across both indoor and outdoor settings.

Roles & Responsibilities:

- Developed the coverage-completeness algorithm that decides in real time during flight whether every surface of a structure has been captured, so the drone keeps scanning until the resulting 3D reconstruction has no gaps.
- Contributed to the vision-based navigation stack that lets the drone localize and move through GPS-denied environments from camera input alone.
- Built the photogrammetry and 3D reconstruction pipeline that turns captured imagery into measurable, photorealistic digital twins for downstream structural inspection.

AI & ML engineer

01/2025 – 06/2026

Limasoft india 🌐

ChartLima:

Natural-language query platform that lets supply-chain analysts ask questions in plain English and get filtered data, analysis, and forecasts, powered by a fine-tuned LLM that generates SQL and routes to modular analysis functions.

Roles & Responsibilities:

- Built an end-to-end NLQ engine by fine-tuning an LLM on domain-specific supply-chain queries to generate SQL for data filtering and structured function calls for routing to the correct analysis module.
- Designed a function-registry and executor architecture that maps LLM output to decoupled modules, each receiving a pre-filtered dataframe and params dict, keeping analysis logic fully independent of the query layer.
- Built a supplier performance intelligence module of 50+ custom functions covering OTD, lead-time and delay analysis, price variance, cost competitiveness, risk assessment, tier classification, and automated scorecard generation.

DS Pipeline

Built drag-and-drop based ML pipelines within a no-code/low-code data science product. Special focus on end-to-end pipeline planning, method implementation, and integration.

Roles & Responsibilities:

- Designed and built a modular drag-and-drop ML pipeline system for a no-code/low-code data science platform, enabling users to build end-to-end ML workflows without coding.
- Developed core pipeline components including preprocessing, feature engineering, data validation, classical ML models, and result evaluation modules.
- Built metadata generation logic to enable real-time UI configuration, validation, and execution traceability across the platform.

Machine Learning Engineer

Kotai Electronics Pvt. Ltd &

11/2021 – 12/2023

Kolkata, India

Automatic Traffic Counter and Classifier(Product)

Description:

Automatic Traffic Counter and Classifier(ATCC) is a device used for traffic volume counting surveys and classifying vehicles into different Vehicle categories and generating reports in an Excel sheet.

Roles &Responsibilities:

- Collected and curated a dataset of over 10,000 + custom traffic images for model training.
- Annotated and validated the dataset to ensure accurate labeling, using tools like LabelImg and data validation scripts.
- Developed and fine-tuned a deep learning model for vehicle classification using Python, Computer Vision,PyTorch
- Implemented data augmentation techniques to improve model generalization.
- Evaluated model performance with metrics like accuracy, precision, recall, and F1 score using **scikit-learn** .
- Deployed the trained model using Flask for real-time vehicle counting and classification.

QA Engineer

08/2020 – 10/2021

Kolkata, India

- Writing and Executing Test Cases and Scenarios: Proficient in creating and executing thorough test cases and scenarios for software applications
- SDLC (Software Development Life Cycle): Experienced in working through the entire software development life cycle,
- Agile Testing: Skilled in Agile testing methodologies, actively participating in sprint planning, conducting continuous testing,

TECHNICAL PROFICIENCY

Languages

Python , HTML , CSS , SQL

AI / ML

Machine Learning, Deep Learning , GenAI, Agentic AI, Time series, RAG

Libraries

Pandas, NumPy ,Scikit-Learn , SKTime,TSA

Databases

MySQL,Postgresql, MongoDB

Frameworks

Flask, LangGraph, PyTorch , TensorFlow

MLOps

Git, Docker, Azure

EDUCATION

Master of Computer Application

krupajal engineering college

Earned a master's degree in Computer Science with a CGPA of **8.2**

08/2023 – 05/2025

Bhubaneswar, India

Bachelor of Computer Applications

Trident Academy of technology

Earned a bachelor's degree in Computer Science with a CGPA of **7.6**

08/2017 – 01/2021

Bhubaneswar, India

LANGUAGES

- English
- Hindi
- Odia